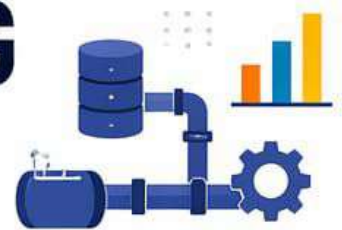




DATA ENGINEERING ROADMAP



STEP BY STEP GUIDE TO BECOME A DATA ENGINEER

LEARN → PRACTICE → BUILD → GROW

1



Programming Basics

- Python Fundamentals
- Core Concepts: Variables, Loops, Functions
- Data Structures
- OOP Concepts (Recommended)

2-4 Weeks

2



SQL & Databases

- SQL Basics to Advanced
- Joins, Subqueries, CTEs, Window Functions
- Normalization vs Denormalization
- Indexing & Query Optimization

2-4 Weeks

3



Data Modeling

- Understanding Data Models
- ER Diagrams
- Dimensional Modeling (Star & Snowflake Schema)
- Fact & Dimension Tables

2-3 Weeks

4



Data Warehousing Concepts

- What is a Data Warehouse?
- OLTP vs OLAP
- Data Marts
- ETL vs ELT

2-3 Weeks

5



ETL/ELT Fundamentals

- Extract, Transform, Load Explained
- ETL Design Patterns
- Data Quality & Validation
- Handling Slowly Changing Dimensions

3-4 Weeks

6



Python for Data Engineering

- Pandas, NumPy
- File Handling (CSV, JSON, Parquet)
- Data Processing
- Automation with Python

3-4 Weeks

7



Big Data Technologies

- Hadoop Ecosystem (HDFS, YARN)
- Apache Spark (Core, SQL, DataFrames)
- PySpark Basics
- When to Use Big Data Tools

4-6 Weeks

8

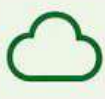


Streaming & Real-Time Data

- Introduction to Kafka
- Kafka Producers & Consumers
- Stream Processing Concepts
- Use Cases

2-3 Weeks

9



Cloud Platforms

- AWS (S3, EC2, Glue, Redshift)
- Or Azure (Data Factory, Synapse)
- Or GCP (BigQuery, Dataflow, GCS)
- Cloud Storage & Compute Basics

3-4 Weeks

10



Workflow Orchestration

- Introduction to Apache Airflow
- DAGs, Tasks, Operators
- Scheduling & Monitoring
- Best Practices

2-3 Weeks

11



Data Quality, Testing & Monitoring

- Data Validation Techniques
- Great Expectations Basics
- Logging & Alerting
- Monitoring Pipelines

2-3 Weeks

12



Projects & Portfolio Building

- End-to-End Data Pipeline Project
- Work with Real Datasets
- Deploy on Cloud
- Document & Showcase Projects

Ongoing

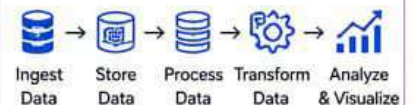
ESSENTIAL SKILLS

- SQL & Databases
- Programming (Python)
- Data Modeling
- ETL/ELT
- Big Data Technologies
- Cloud Platforms
- Problem Solving
- Automation & Scripting

IMPORTANT TOOLS

- Python
- SQL
- Pandas
- Spark
- Hadoop
- Kafka
- Airflow
- AWS
- Azure
- GCP
- Docker
- Git

DATA ENGINEERING LEARNING PATH



PROJECT IDEAS

- End-to-End ETL Pipeline (Batch)
- Real-Time Data Pipeline with Kafka & Spark
- Data Warehouse Project on AWS/Azure/GCP
- Data Lake with Medallion Architecture
- Pipeline Monitoring Dashboard

CAREER PATHS

- Data Engineer
- Big Data Engineer
- ETL Developer
- Cloud Data Engineer
- Data Platform Engineer